BSc Applied Computing
TRUR3120 Professional Project
Project Proposal & Ethical Approval Form

1. Research details	
Student Name/s: Max Walker	**Supervisor:** John Glazebrook, Richard Morris
Programme title:	**Module Title:**
Project Title: Educational computer game simulation of an 18th century Cornish mining community.	

Please complete this form along with Form A, B or C as appropriate. Failure to give sufficient information under each section will delay the approval process. Any additional information may be provided on additional sheets of paper.

2. Aims and objectives - *Describe the overall aim of the project and at least 4 specific, measurable, achievable, relevant and time bound objectives.*

To apply my skills of software development, project management, data analysis, and systems thinking to design and produce a complex video game simulation of an 18th century Cornish mining community. Ultimately to raise awareness and understanding of the globally transformative process of industrialisation that began in the Cornish countryside 250 years ago.

To this end I will need to:

- Research existing works that use software simulations to model processes of industrialisation, urban transition, and other related socio-economic phenomena to inform my own approach.
- Consistently adhere to a flexible, iterative design and development process while applying industry best practices.
- Design and produce a functional and intuitive user interface for the application with Godot.
- Design and develop an interconnected simulation of both a mine and village that can be influenced by player action with Godot.

3. Project risks assessment and feasibility study - *Analyse the possible risks to the project achieving its objectives and actions to mitigate those risks.*

The single greatest risk involved with this project is scope creep. Video games are enormously complex undertakings that are very easy to generate ideas for, and so can quickly spiral into infeasibility without proper attention given to managing project goals.

A flexible but goal-oriented Agile approach with regular standups and reviews will greatly mitigate risks of scope creep, allowing me to continuously assess the velocity and projected complexity of the project and adjust my process accordingly.

4. **Technical requirements -** *Describe any technical resources such as hardware, software, or network connections needed by the project. Indicate any that may require installation or special permissions by Truro and Penwith college. Indicate any that might pose a security risk to the college.*

The development process should require just my own hardware, and I will use the open-source game engine Godot so no software licences should be necessary.

5. **External stakeholders -** *Are any outside organisations or individuals involved in the project?*

Two of my personal friends have given me verbal confirmation that they'd produce artwork for this project.

6. **Methodology, Analysis and Design -** *Describe the project methodology you intend to follow and include any initial analysis and design information.*

I intend to follow an Agile framework loosely structured on Scrum to ft my needs as a solo developer. I will use the Trello Kanban tool to both plan and manage my progress with weekly sprints and periodic standups.

The game will be designed following a model-view-controller pattern to separate out concerns and ensure the project remains modular and frictionless to maintain or extend, making use of interfaces between different domains

7. **Research & References Sources**

- V Lehdonvirta, E Castronova, 2014. Virtual Economies: Design and Analysis
- N Visonà, L Riccetti, 2024. Simulating the industrial revolution: a history-friendly model
- J Ge, BA Furtado, 2023. Modelling urban transition with coupled housing and labour markets
- L Rosenburger, et al, 2026. Innovation Networks in the Industrial Revolution
- B Ma, 2022. System Theory and Cybernetics of the British Industrial Revolution: Positive Feedback Between Smith-Young Growth and Schumpeterian Growth

8. Ethical and Higher level clearance

Please indicate whether your research intends to include any of the following aspects:

	Yes	No
Use of Human Participants *Including surveys and storing of personal data. If so please complete Form A below*	☐	☐
Use of Non-Human Participants *e.g. animals, If so please complete Form B below*	☐	☐
Use of Non-living Human Participants *If so, please complete Form C below*	☐	☐

Higher level clearance.

Please identify whether your research intends to include any of the following aspects:

	Yes	**No**
• Active research with children	☐	☐
• Ingestion	☐	☐
• Invasive studies (ie. blood samples)	☐	☐
• Active research with vertebrates	☐	☐
• Research involving non-living human participants	☐	☐

If any of these aspects are being studied please refer to the *Notes for Guidance* in order to identify what supplementary paperwork must be included.

9. Signatures

9.a Supervisor

I have discussed the design of this research project with the applicant.

Lecturer Signed:_____ dated: _____

Supervisor Signed:_____ dated: _____

9.b Programme Level Ethics Review Panel Approval

We agree that this project has been:

- ☐ referred to College Ethics Board
- ☐ approved at Programme Level and the research may continue
- ☐ approved at Programme Level and the research may continue subject to completion of the agreed action points and/or clarification*

Signed:_____ Programme/Module Leader

Signed:_____ Critical friend

9.c College Ethics Board Approval

We agree that this project has been:

- ☐ approved at the College Ethics Board and the research may continue
- ☐ approved at the College Ethics Board and the research may continue subject to completion of the agreed action points*
- ☐ declined at the College Ethics Board. Review of methodology recommended.

Signed:_____ Ethics Board Chair or nominee

Date of Board:_____

* Attach agreed action points to the signed form

Form A – Use of Human Participants

Ethical procedures Please indicate <u>how</u> you will ensure that your research conforms with each clause of the University of Plymouth's *Ethical Principles for Research Involving Human Participants.*
Informed consent
Openness and honesty
Right to withdraw
Protection from harm
Debriefing
Confidentiality
Other professional bodies

Form B – Use of Non-Human Participants

Ethical procedures Please indicate <u>how</u> you will ensure that your research conforms with each clause of the University of Plymouth's *Ethical Principles for Research Involving Non-Human Participants.*
Legal and moral responsibilities
Avoidance of unnecessary study
Environmental impacts
Monitoring
Justifiable benefits

Form C – Use of Non-living Human Participants

Ethical procedures Please indicate <u>how</u> you will ensure that your research conforms with each clause of the University of Plymouth's *Ethical Principles for Research Involving Non-Living Human Participants.*
Respect for the dead
Permission from guardians of the dead
Scientific considerations
Negotiated agreements